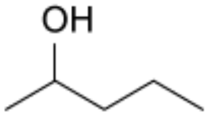
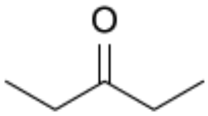
Which of the following compounds can be used to directly synthesize a compound containing a five-carbon alkyl chain with a cyanohydrin at carbon 2?

○ A.



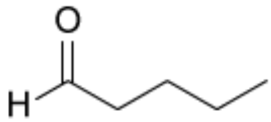
[31%]

○ B.



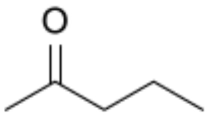
[2%]

○ C.



[12%]

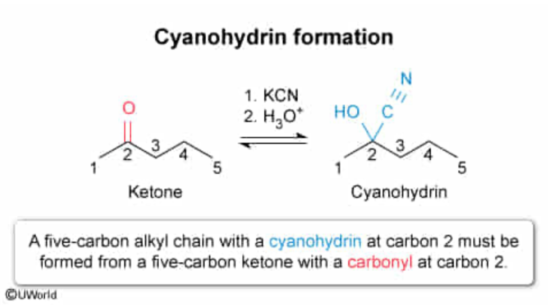
✔ ◉ D.



[53%]

Correct
53% answered correctly

Explanation:



Ketones and aldehydes can be transformed into other functional groups via nucleophilic addition to their carbonyl. In a nucleophilic addition, a nucleophile attacks the electrophilic carbonyl carbon and forms a new sigma bond, and the pi bond of the carbonyl breaks to form an alkoxide. A cyanohydrin is a functional group combination where a **hydroxyl group** (–OH) and a **cyano group** (–C≡N) are attached to the **same carbon**. Cyanohydrins are formed by **nucleophilic addition of the cyanide anion (CN^-)** to the carbonyl of a ketone or aldehyde. The resulting alkoxide is **protonated** by an acidic workup.

Because cyanohydrins are formed through the nucleophilic addition of cyanide to a carbonyl, the compound used to form the cyanohydrin described in this question must contain a carbonyl. The question states a five-carbon alkyl chain contains a cyanohydrin group at carbon 2. The starting material must be a ketone because the cyanohydrin is not located at the end of the chain. Therefore, the carbonyl of the ketone should also be at carbon 2. The structure shown in Choice D is a **five-carbon ketone** with a **carbonyl at carbon 2**.

(Choice A) This structure is a secondary alcohol, and therefore does *not* contain the electrophilic carbonyl necessary for cyanohydrin formation through nucleophilic addition.

(Choice B) Although this structure is a five-carbon ketone, its carbonyl is at carbon 3 instead of carbon 2. Therefore, the cyanohydrin forms at the incorrect carbon and does not give the desired product.

(Choice C) This structure is a five-carbon aldehyde. Because an aldehyde inherently has its carbonyl at the end of a chain, conversion of an aldehyde to a cyanohydrin yields a cyanohydrin at *carbon 1* instead of carbon 2.

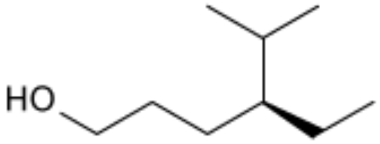
Educational objective:
A cyanohydrin is a functional group combination where a hydroxyl group and a cyano group are attached to the same carbon. Cyanohydrins are generated by the nucleophilic addition of a cyanide ion to the carbonyl of a ketone or aldehyde, followed by an acidic workup.

Time spent:0 sec
Subject:
Organic Chemistry

QID:404179
System:
Functional Groups and Their Reactions

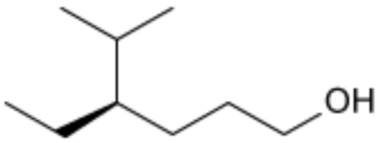
An alcohol is oxidized with $KMnO_4$ to (4*R*)-4-ethyl-5-methylhexanoic acid. What is the structure of the alcohol?

○ A.



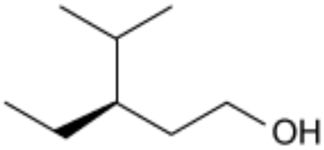
[38%]

✔ ◉ B.



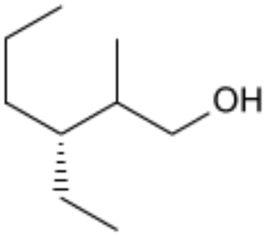
[43%]

○ C.



[3%]

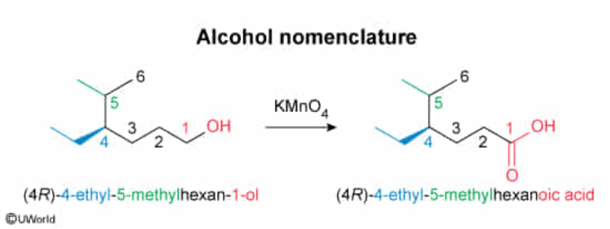
○ D.



[14%]

Correct
43% answered correctly

Explanation:



Alcohols are molecules containing a **hydroxyl group** (–OH) and are named using systematic rules from the IUPAC. When an alcohol is the highest-priority functional group, the parent chain is the longest continuous carbon chain containing the alcohol. The parent chain prefix corresponds to the number of carbon atoms, and the suffix –*ol* corresponds to the alcohol functional group. The parent chain is numbered to give the hydroxyl group the **lowest number**, and its location is indicated right before the suffix. Stereocenters are **listed at the beginning** of the compound name in parentheses followed by **substituents**, each using a number to designate its position on the parent chain.

The question states that an alcohol is oxidized with $KMnO_4$ to (4*R*)-4-ethyl-5-methylhexanoic acid. Because primary alcohols are the only type of alcohol that can be oxidized to carboxylic acids and alkyl chains do not react with $KMnO_4$, the solution must be **a primary alcohol with the same parent chain length and substituent pattern**. The prefix *hex*– indicates six carbons are present in the parent chain. An ethyl substituent with an *R* configuration must also exist at carbon 4, and a methyl substituent at carbon 5.

Therefore, the IUPAC name of the alcohol is (4*R*)-4-ethyl-5-methylhexan-1-ol, which is depicted by the structure shown in Choice B.

(Choice A) Carbon 4 in this structure has an *S* configuration. Therefore, this structure is the enantiomer of the oxidized alcohol.

(Choice C) The parent chain in this structure contains five carbons (one carbon *shorter* than the carboxylic acid). During an alcohol oxidation, the *carbon* atom attached to the hydroxyl is oxidized, so the parent chain of the alcohol and carboxylic acid must contain the *same number of carbons*.

(Choice D) This structure is obtained if the parent chain is numbered incorrectly to give hydroxyl the highest location position.

Educational objective:
Alcohols are molecules containing a hydroxyl group and are named using the suffix –*ol*. The prefix indicates the length of the parent chain, and the position of the hydroxyl is indicated using its position number before the suffix. Stereocenters and substituents are listed preceding the parent chain name.

Time spent:0 sec
Subject:
Organic Chemistry

QID:404180
System:
Functional Groups and Their Reactions